Qing Hou

University of Florida, Gainesville | houqing@ufl.edu | +1 2137096185
Ph.D. Candidate in Urban and Regional Planning
Research Interests: Urban Planning AI; Urban Analytics; Transportation and Urban Planning; Landscape Architecture, Physical Planning, Planning & Design Studio, Autonomous Vehicles Acceptance; Public Transportation; Smart, Equitable & Sustainable Mobility; Big Data Analytics; Machine Learning & Deep Learning; Generative AI
Google Scholar | LinkedIn | Personal Website

EDUCATION

University of Florida **Gainesville, Florida, USA**

- Ph.D. Candidate: Urban and Regional Planning, GPA: 3.80/4.00 08/2023 – Now
- Advisor: Prof. Zhong-Ren Peng
- Dissertation Title: Rethinking Public Transit Access: From Trip-Referenced Serviceability and User Perceptions to AI-Enabled Transit Planning and Operations

University of Florida **Gainesville, Florida, USA**

- Master of Science: Electrical and Computer Engineering, GPA: 3.80/4.00 08/2023 – 06/2026
- Selected Coursework: Foundation of Machine Learning, Image Processing and Computer Vision, Pattern recognition and Intelligent Vision, Physics Informed Machine Learning, Introduction to Quantum Computing, IoT Security and Privacy, Computer Communications

University of Southern California **Los Angeles, California, USA**

- Master of Urban Planning, GPA: 3.80/4.00 09/2020 – 06/2022
- Selected Coursework: Physical Planning, Planning Studio, Urban and Community Design, Design Skills for Urban Planners, Planning Theory and History, Planning Law, Housing and Land Use, Capstone in Urban Planning

Northeast Forestry University **Harbin, Heilongjiang, China**

- Bachelor of Engineering: Landscape Architecture, GPA: 3.83/4.00 09/2016 – 06/2020
- Selected Coursework: Landscape Planning and Designing, Landscape Architecture Design, Sketch, Watercolor Landscape Painting, Landscape Floriculture & Dendrology, Landscape Heritage Conservation and Management

JOUNRAL PAPER PUBLICATIONS/ MANUSCRIPTS

Publications

J1. **Hou, Q**., Hafiz, D., Peng, Z. R., & Lu, K. F. (2026). Governing Fully Autonomous Vehicle Adoption: A State-of-the-art Review of Public Acceptance, System Impacts, and Policy Pathways. *Transport Policy*, 104391. https://doi.org/10.1016/j.tranpol.2026.104391

J2. Peng, Z., Lu, K., Liu, Y., **Hou, Q**., Zhang, Q. (2026). Symbiotic Planning Theory: The CORE Framework for Human-AI Cocreation in Urban Planning. *Journal of the American Planning Association*, 1-16. https://doi.org/10.1080/01944363.2026.2640038

J3. Hafiz, D., **Hou, Q**., Peng, Z., Zohdy, I. (2025). Advancing Sustainable Urban Mobility: Public Acceptance and Perceived Risks of Autonomous Vehicle Deployment in Dubai. *Sustainability,* 2025, 17(24), 11021. https://doi.org/10.3390/su172411021

J4. Liu, Y., Peng, Z., **Hou, Q**., & Lu, K. (2024). The Role, Opportunities, and Challenges of Generative AI in Comprehensive Planning of American Small Towns – Using ChatGPT as an Example (in Chinese). *Journal of Urban Regional Planning*, 16(1), 215-228.

J5. **Hou, Q**., Zhou, S., Chi, G. (2024). The Factors Affecting Electric Vehicle Adoption in the United States, 2016-2021. *Proceedings of the 7th International Conference on Economic Management and Green Development*. ICEMGD 2023. Applied Economics and Policy Studies. Springer, Singapore. https://doi.org/10.1007/978-981-97-0523-8_28

J6. Peng, Z., Liu, Y., **Hou, Q**., Lu, K. (2023). A Synthesis of Economic Impact Assessment of Transit Services. *Florida*

Department of Transportation. https://rosap.ntl.bts.gov/view/dot/84601

Manuscripts in Review

J7. **Hou, Q**., Lu, K., Zhang, Q., Peng, Z. (2026). Trip-Referenced Transit Serviceability: Diagnosing Access, Connectivity, and Schedule Gaps in Downtown Tampa. *Journal of Transport Geography*. (Under Review)

J8. **Hou, Q**., Lu, K., Chi, G., Zhang, Q., Peng, Z. (2026). Planning Equitable Urban Park Access: Mobility Dependence, Facility Condition, and Investment Priorities in Los Angeles. *Cities*. (Under Review)

J9. Lu, K., **Hou, Q**., Zhang, Q., Liu, Y., Peng, Z. (2026). Planning Automation of Shared Micromobility System Design: A Multi-Agent Deep Reinforcement Learning Approach. *Nature Computational Science*. (Under Review)

J10. Lu, K., **Hou, Q**., Andrews, J., Nejat, A., Ewing, B. (2026). Financial Implementation Timing and Administrative Capacity in U.S. CDBG-DR Recovery. *Nature Cities*. (Under Review)

J11. Liu, Y., Lu, K., Zhang, Q., **Hou, Q**., Peng, Z. (2026). Are Current Streets Suitable for Autonomous Bus Transit? User Perception and Street View Imagery Analysis across 12 U.S. Sites. *Sustainable Cities and Society*. (Under Review)

J12. Lu, K., **Hou, Q**., Peng, Z. (2026). Impacts of Shared Micromobility System Size on Usage Patterns: Planning Insights. *Journal of Planning Education and Research*. (Under Review)

J13. Lu, K., **Hou, Q**., Peng, Z. (2026). Highway Traffic Forecasting and Resilience Assessment During Hurricane Dorian Using Multi-Graph Learning. *Transportation Research Part D: Transport and Environment*. (Under Review)

J14. Liu, Y., Lu, K., **Hou, Q**., et al. (2026). Can Discounted Uber Rides Connecting to Transit Help Regain Bus Ridership? Evidence from Pinellas County, Florida. *Transport Policy*. (Under the Second Round of Review)

J15. Lu, K., **Hou, Q**., Liu, Y., Peng, Z. (2026). Shared Micromobility and Public Transit Integration in Florida Cities: Network Accessibility and Feeder-Mode Behavior. *Travel Behavior and Society*. (Under Review)

J16. Lu, K., **Hou, Q**., Andrews, J., Nejat, A. (2026). Measuring Governmental CDBG-DR Fund Implementation Capacity in Disaster Recovery: Evidence from Hurricane Harvey in Texas. *International Journal of Disaster Risk Reduction*. (Under Review)

J17. Lu, K., **Hou, Q**., Popoola, T., Hayhoe, K., Andrews, J., Nejat, A. (2026). Housing-Stock Inequalities and the Timing of Future Growth: Climate-Housing Exposure in Harris County, Texas. *Urban Climate*. (Under Review)

J18. Zhang, Q., Peng, Z., **Hou, Q**., Lu, K., Wang, D., Liu, Y. (2026). Human-AI Collaboration in Transit Network Design: A Data-Driven Planning Case from Tampa. *Journal of the American Planning Association*. (Under Review)

J19. Lu, K., **Hou, Q**., Popoola, T., Hayhoe, K., Andrews, J., Nejat, A. (2026). Wildfire Risk Assessment in the United States: A State-of-the-Art Review and Integrated Framework for Climate-Informed Decision-Making. *Risk Analysis*. (Under Review)

RESEARCH EXPERIENCE

SaferEast Initiative: Causal Evaluation of AI-Based CPTED Interventions in East Gainesville, FL 04/2026-09/2026

- Funded by Arnold Ventures Arnold Ventures at the University of Florida, $91.4k
- Conducted a comprehensive literature review on Crime Prevention Through Environmental Design (CPTED) and developed a community survey questionnaire assessing perceived safety, environmental conditions, and public acceptance of AI-assisted safety technologies.
- Developed a conceptual framework and implementation roadmap for an AI-driven public safety dashboard, integrating spatiotemporal crime analysis, predictive modeling, data governance, and decision-support functions. Survey.

Transit Comparative Analysis and 2050 Development Forecast for Central Florida 07/2024-02/2026

- Funded by Florida Department of Transportation, $388k
- Outlined a comprehensive methodology to analyze FDOT District Five's (FDOT D5) existing transit systems, forecast future transit needs, and present a potential roadmap to meeting those future needs using lessons learned from equivalent regions/transit systems worldwide and emerging transportation technology solutions.

User Perception for Autonomous Vehicle Deployment: Dubai Case Study 10/2023-07/2024

- Funded by Ajman University Research Grant
- Summarized and analyzed interview data using natural language processing (NLP) and the GPT API, including sentiment analysis; authored the corresponding analysis section and the manuscript introduction section.

A Synthesis of Economic Impact Assessment of Transit Services 07/2023-11/2023

- Funded by Florida Department of Transportation, $150k, available at https://rosap.ntl.bts.gov/view/dot/84601
- Conducted in-depth synthesis of the economic impact assessment of transit services in the United States from 2003 to 2023, focusing on Benefit-Cost Analysis (BCA) and Economic Impact Analysis (EIA) methodologies, examining their application in public transit systems, and highlighting geographical variations and trends.

The Factors Affecting Electric Vehicle Adoption in the United States, 2016-2021 09/2022-07/2023

- Funded by USDA National Institute of Food and Agriculture and Multistate Research Project #PEN04623 and the Eunice Kennedy Shriver National Institute of Child Health and Human Development.
- Applying ordinary least squares (OLS) and machine learning (ML) models to explore the relationships between external factors and electric vehicle (EV) adoption using state-level data in the contiguous United States from 2016 to 2021.
- Published at Proceedings of the 7th International Conference on Economic Management and Green Development.

Inequity of Park Accessibility 09/2022-07/2023

- Funded by USDA National Institute of Food and Agriculture and Multistate Research Project #PEN04623 and the Eunice Kennedy Shriver National Institute of Child Health and Human Development.
- Proposed an integrated park accessibility approach combining subjective and objective perspectives and examined its feasibility by comparing it with traditional container methods in the city of Los Angeles.

DESIGN EXPERIENCE

River Link: Mixed-Use Riverfront Redevelopment, Los Angeles, California 05/2022

- Developed a riverfront master plan integrating 500 affordable housing units, mixed-use development, and public green spaces.
- Proposed multimodal transportation improvements, including light-rail station relocation, pedestrian and bicycle connections, and complete streets, while considering environmental remediation and implementation costs.
- https://github.com/qinghou1/Design_Portfolio/blob/main/USC_Capstone_2.pdf

Green Network: Taylor Yard G2 Riverfront Revitalization, Los Angeles, California 04/2022

- Analyzed park accessibility, environmental burdens, and neighborhood socioeconomic conditions to inform riverfront revitalization strategies.
- Designed an interconnected green-space network incorporating pedestrian bridges, bicycle paths, community spaces, and flood-resilient landscape features.
- https://github.com/qinghou1/Design_Portfolio/blob/main/USC_Capstone.pdf

Mixed-Use Affordable Housing Master Plan, Oxnard, California 05/2021

- Developed a 62-acre mixed-use master plan integrating diverse housing options, commercial spaces, and recreational facilities for families, seniors, and local workers.
- Evaluated transportation accessibility, zoning constraints, and surrounding land uses to design connected pedestrian and bicycle networks, shared green spaces, and community facilities.
- https://github.com/qinghou1/Design_Portfolio/blob/main/USC_Mixed_Use_Affordable_Housing.pdf

Filtering Network: Adaptive Shoreline Design for the Mississippi River Delta 06/2020

- Developed an adaptive shoreline design strategy integrating ecological restoration, coastal protection, and sustainable landscape development.
- Designed an ecological filtration network to improve soil and water quality, restore wetlands, and repurpose degraded infrastructure and landscapes.

- https://github.com/qinghou1/Design_Portfolio/blob/main/Filtering_Network.pdf

Urban Living Room: Ecological Landscape Design at Songhua Lake 11/2019

- Designed a 34-hectare island landscape master plan promoting outdoor recreation, social interaction, and connections with nature among online gaming communities.
- Integrated permeable paving, layered vegetation, and adaptive shoreline treatments to address water pollution and seasonal water-level fluctuations.
- https://github.com/qinghou1/Design_Portfolio/blob/main/Urban_Living_Room.pdf

CONFERENCE PRESENTATIONS

C1. **2027.01: Transportation Research Board 106th Annual Meeting.** *Shared Micromobility and Public Transit Integration in Florida Cities: Network Accessibility and Feeder-Mode Behavior*. Washington, D.C., USA.

C2. **2027.01: Transportation Research Board 106th Annual Meeting.** *Impacts of Shared Micromobility System Size on Usage Patterns: Planning Insights*. Washington, D.C., USA.

C3. **2027.01: Transportation Research Board 106th Annual Meeting.** *From Transit Coverage to Trip-Referenced Serviceability: Spatial, Topological, and Schedule Gaps in Downtown Tampa*. Washington, D.C., USA.

C4. **2026.10: Association of Collegiate Schools of Planning (ACSP) Conference**. *Data-Driven Transit Network Design Through Hierarchical Reinforcement Learning A Multi-Agent Framework for Route and Station Planning*. Pittsburgh, PA, USA.

C5. **2025.10: Association of Collegiate Schools of Planning (ACSP) Conference**. *Evaluating Transit Service Gaps Through Empirical Travel Patterns from Downtown Tampa, Florida*. Minneapolis, MN, USA.

TEACHING EXPERIENCE

Teaching Assistant | University of Florida, *Planning for Climate Change (Graduate level)* Fall 2026
Teaching Assistant | University of Florida, *Transportation and Land Use Modeling (Graduate level)* Spring 2025
Teaching Assistant | Northeast Forestry University, *Planning and Design Studio (Undergraduate level)* Spring 2019&2020

PROFESSIONAL EXPERIENCE

Research Assistant | Advisor: Dr. Zhong-Ren Peng, Planning, University of Florida 08/2023 – Now
Volunteer Research | Advisor: Dr. Guangqing Chi, University of Pennsylvania State University 07/2022 – 08/2023
Summer Research Intern | Advisor: Dr. Stanislav Sobolevsky, New York University 08/2021 – 09/2021
Planning Intern | Shanghai Tongji Urban Planning & Design Institute 05/2021 – 08/2021

ACADEMIC SERVICES

ACSP Planning Knowledge Hub

- Integrate existing urban planning knowledge and resources particularly related to Urban Planning AI
- Develop official website: https://qinghou1.github.io/ACSP_Planning_Knowledge_Hub/

UF Master of Science & Graduate Certificate in Urban Analytics Program

- Design flyers and brochures for advertising master degree and graduate certificate in urban analytics
- Develop official website: https://qinghou1.github.io/MSUA/

Affiliations

- International Center for Adaptation Planning and Design (iAdapt), Member

Conference Events

- Co-hosted "*AI and Cities: An International Forum for Innovation and Collaboration*" at the University of Florida, 2025 & 2027 (upcoming event)

COMMUNITY OUTREACH

UF Research Expo 09/23/2026

- Engaged with UF undergraduate communities to share iAdapt research center's work and advertise urban and regional planning department's new master degree and graduate certificate in urban analytics

The 27th Annual Stop the Violence Back-to-School Rally 07/25/2026

- Supported community outreach for the SaferEast Initiative by distributing survey recruitment brochures, raising awareness of the research, and encouraging local residents to participate in a community safety survey on retail parking lot safety and crime prevention

HONORS & AWARDS

Best Presentation/Research Award, AI and Cities: An International Forum for Innovation and Collaboration, hosted by the College of Design, Construction and Planning, University of Florida, USA 2025
Outstanding Undergraduate, Northeast Forestry University (Top 3%) 2020
Second-class Scholarships of Northeast Forestry University (2017-2019) (top 10%) 2017-2019
Third Price of the Fourth Garden Cup, China Society of Landscape Architecture 2018
Merit Student, Northeast Forestry University 2018
Third Prize of Provincial Construction Festival, Landscape Architecture Association in Heilongjiang Province 2017
First-class Scholarships of Northeast Forestry University (top 5%) 2016

SKILLS

Toolkit: Python, SPSS, PTV VISUM, ArcGIS Pro, AutoCAD, Photoshop, Illustrator, InDesign, SketchUp, Lumion
Skillset: Geospatial Analysis, Data Visualization, Machine Learning, Natural Language Processing, Programming
Language: English (TOEFL: 110), Mandarin (Native)

REFERENCES

Zhong-Ren Peng
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Emre Tepe
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Yanghe Liu
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